

LISTA DE MAT00A

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Técnicas de derivação

3–32 Derive a função.

3. $f(x) = 186,5$

5. $f(x) = 5x - 1$

7. $f(x) = x^3 - 4x + 6$

9. $f(t) = \frac{1}{4}(t^4 + 8)$

11. $y = x^{-2/5}$

13. $V(r) = \frac{4}{3}\pi r^3$

15. $Y(t) = 6t^{-9}$

17. $G(x) = \sqrt{x} - 2e^x$

19. $F(x) = \left(\frac{1}{2}x\right)^5$

21. $y = ax^2 + bx + c$

23. $y = \frac{x^2 + 4x + 3}{\sqrt{x}}$

25. $y = 4\pi^2$

27. $H(x) = (x + x^{-1})^3$

29. $u = \sqrt[3]{t} + 4\sqrt{t}$

31. $z = \frac{A}{y^{10}} + Be^y$

4. $f(x) = \sqrt{30}$

6. $F(x) = -4x^{10}$

8. $f(t) = \frac{1}{2}t^6 - 3t^4 + t$

10. $h(x) = (x - 2)(2x + 3)$

12. $y = 5e^x + 3$

14. $R(t) = 5t^{-3/5}$

16. $R(x) = \frac{\sqrt{10}}{x^7}$

18. $y = \sqrt[3]{x}$

20. $f(t) = \sqrt{t} - \frac{1}{\sqrt{t}}$

22. $y = \sqrt{x}(x - 1)$

24. $y = \frac{x^2 - 2\sqrt{x}}{x}$

26. $g(u) = \sqrt{2}u + \sqrt{3}u$

28. $y = ae^v + \frac{b}{v} + \frac{c}{v^2}$

30. $v = \left(\sqrt{x} + \frac{1}{\sqrt[3]{x}}\right)^2$

32. $y = e^{x+1} + 1$

3–26 Derive.

3. $f(x) = x^2 e^x$

5. $y = \frac{e^x}{x^2}$

4. $g(x) = \sqrt{x} e^x$

6. $y = \frac{e^x}{1+x}$

$$7. \quad g(x) = \frac{3x - 1}{2x + 1}$$

$$8. \quad f(t) = \frac{2t}{4 + t^2}$$

$$9. \quad V(x) = (2x^3 + 3)(x^4 - 2x)$$

$$10. \quad Y(u) = (u^{-2} + u^{-3})(u^5 - 2u^2)$$

$$11. \quad F(y) = \left(\frac{1}{y^2} - \frac{3}{y^4} \right) (y + 5y^3)$$

$$12. \quad R(t) = (t + e^t)(3 - \sqrt{t})$$

$$13. \quad y = \frac{x^3}{1 - x^2}$$

$$14. \quad y = \frac{x + 1}{x^3 + x - 2}$$

$$15. \quad y = \frac{t^2}{3t^2 - 2t + 1}$$

$$16. \quad y = \frac{t^3 + t}{t^4 - 2}$$

$$17. \quad y = (r^2 - 2r)e^r$$

$$18. \quad y = \frac{1}{s + ke^t}$$

$$19. \quad y = \frac{v^3 - 2v\sqrt{v}}{v}$$

$$20. \quad z = w^{3/2}(w + ce^w)$$

$$21. \quad f(t) = \frac{2t}{2 + \sqrt{t}}$$

$$22. \quad g(t) = \frac{t - \sqrt{t}}{t^{1/3}}$$

$$23. \quad f(x) = \frac{A}{B + Ce^x}$$

$$24. \quad f(x) = \frac{1 - xe^x}{x + e^x}$$

$$25. \quad f(x) = \frac{x}{x + \frac{c}{x}}$$

$$26. \quad f(x) = \frac{ax + b}{cx + d}$$

1-16 Derive.



1. $f(x) = x - 3 \operatorname{sen} x$

2. $f(x) = x \operatorname{sen} x$

3. $y = \operatorname{sen} x + 10 \operatorname{tg} x$

4. $y = 2 \operatorname{cosec} x + 5 \cos x$



5. $g(t) = t^3 \cos t$

6. $g(t) = 4 \sec t + \operatorname{tg} t$

7. $h(\theta) = \operatorname{cosec} \theta + e^\theta \cotg \theta$

8. $y = e^u (\cos u + cu)$



9. $y = \frac{x}{2 - \operatorname{tg} x}$

10. $y = \frac{1 + \operatorname{sen} x}{x + \cos x}$

11. $f(\theta) = \frac{\sec \theta}{1 + \sec \theta}$

12. $y = \frac{1 - \sec x}{\operatorname{tg} x}$

13. $y = \frac{\operatorname{sen} x}{x^2}$

14. $y = \operatorname{cosec} \theta (\theta + \cotg \theta)$

15. $f(x) = xe^x \operatorname{cosec} x$

16. $y = x^2 \operatorname{sen} x \operatorname{tg} x$

7-46 Encontre a derivada da função.

7. $F(x) = (x^3 + 4x)^7$

8. $F(x) = (x^2 - x + 1)^3$

9. $F(x) = \sqrt[4]{1 + 2x + x^2}$

10. $f(x) = (1 + x^4)^{2/3}$

11. $g(t) = \frac{1}{(t^4 + 1)^3}$

12. $f(t) = \sqrt[3]{1 + \operatorname{tg} t}$

13. $y = \cos(a^3 + x^3)$

14. $y = a^3 + \cos^3 x$

15. $y = xe^{-kx}$

16. $y = 3 \cot(n\theta)$

17. $g(x) = (1 + 4x)^5 (3 + x - x^2)^8$

18. $h(t) = (t^4 - 1)^3 (t^3 + 1)^4$

19. $y = (2x - 5)^4 (8x^2 - 5)^{-3}$

20. $y = (x^2 + 1) \sqrt[3]{x^2 + 2}$

21. $y = \left(\frac{x^2 + 1}{x^2 - 1} \right)^3$

22. $y = e^{-5x} \cos 3x$

23. $y = e^{x \cos x}$

24. $y = 10^{1-x^2}$

25. $F(z) = \sqrt{\frac{z-1}{z+1}}$

26. $G(y) = \frac{(y-1)^4}{(y^2+2y)^5}$

27. $y = \frac{r}{\sqrt{r^2+1}}$

28. $y = \frac{e^{2a}}{e^a + e^{-a}}$

29. $y = \operatorname{tg}(\cos x)$

30. $G(y) = \left(\frac{y^2}{y+1} \right)^5$

31. $y = 2^{\operatorname{sen} \pi x}$

32. $y = \operatorname{tg}^2(3\theta)$

33. $y = \sec^2 x + \operatorname{tg}^2 x$

34. $y = x \operatorname{sen} \frac{1}{x}$

35. $y = \cos \left(\frac{1 - e^{2x}}{1 + e^{2x}} \right)$

36. $F(z) = \sqrt{\frac{t}{t^2+4}}$

2-22 Derive a função.

2. $f(x) = \ln(x^2 + 10)$

3. $f(x) = \sin(\ln x)$

5. $f(x) = \log_2(1 - 3x)$

7. $f(x) = \sqrt[5]{\ln x}$

9. $f(x) = \sqrt{x} \ln x$

11. $F(t) = \ln \frac{(2t + 1)^3}{(3t - 1)^4}$

13. $g(x) = \ln(x \sqrt{x^2 - 1})$

15. $y = \frac{\ln x}{1 + x}$

17. $y = \ln |2 - x - 5x^2|$

19. $y = \ln(e^{-x} + xe^{-x})$

21. $y = 2x \log_{10} \sqrt{x}$

4. $f(x) = \ln(\sin^2 x)$

6. $f(x) = \log_3(xe^x)$

8. $f(x) = \ln \sqrt[5]{x}$

10. $f(t) = \frac{1 + \ln t}{1 - \ln t}$

12. $h(x) = \ln(x + \sqrt{x^2 - 1})$

14. $F(y) = y \ln(1 + e^y)$

16. $y = \ln(x^4 \sin^2 x)$

18. $H(z) = \ln \sqrt{\frac{a^2 - z^2}{a^2 + z^2}}$

20. $y = [\ln(1 + e^x)]^2$

22. $y = \log_2(e^{-x} \cos \pi x)$
